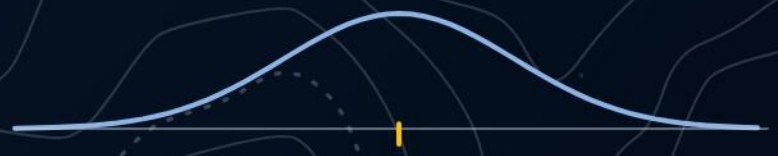
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A KINVECTA SERIES

How Large Language Models Work

From math to transformers

EPISODE 08 · THE LOSS, PART 2



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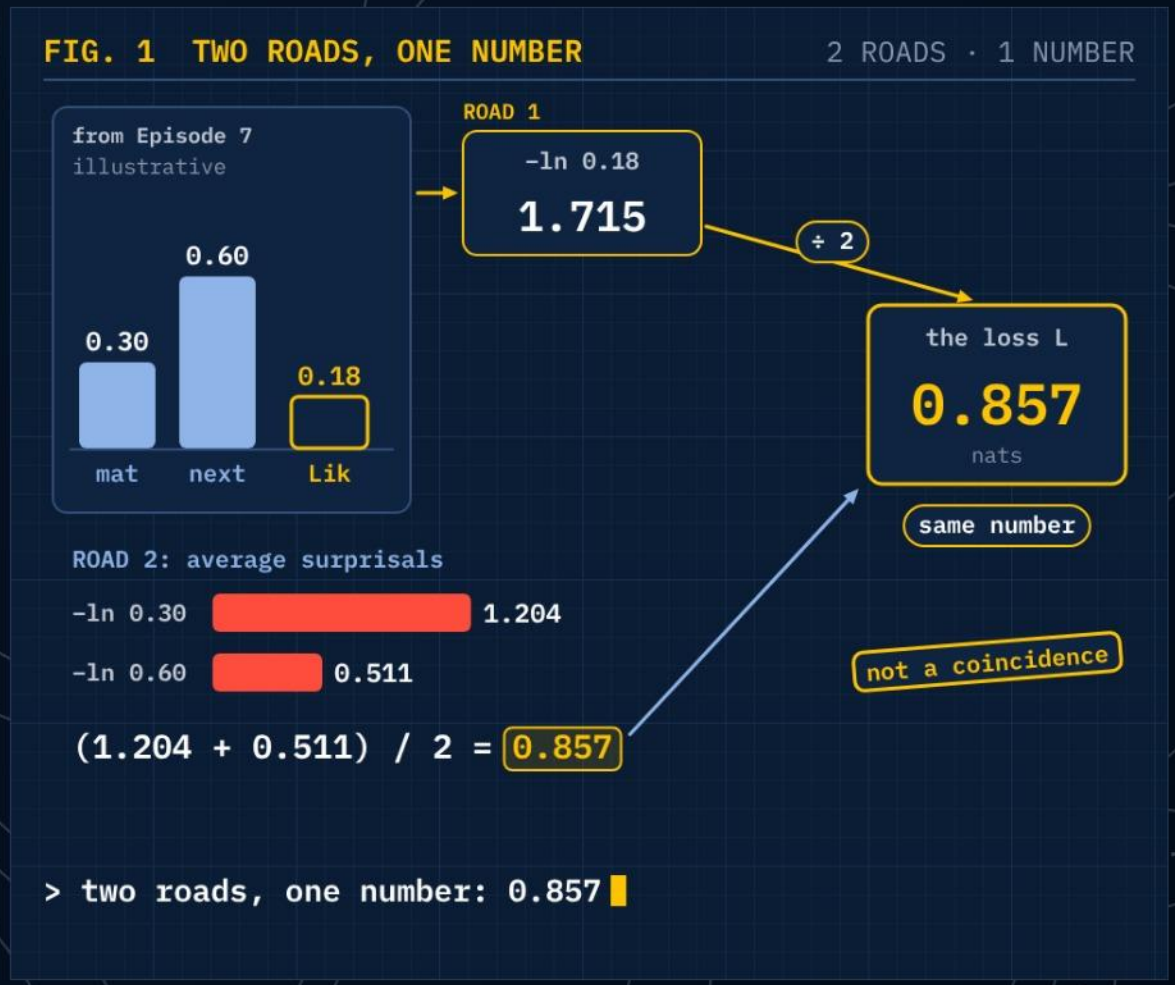
HOW LARGE LANGUAGE MODELS WORK · EPISODE 8 ·

Two roads. One number.

Multiply the probabilities, or average the surprisals. Both land on 0.857. That's not a coincidence.

11-minute lesson

Companion notebook



MARK

WHY THIS MATTERS

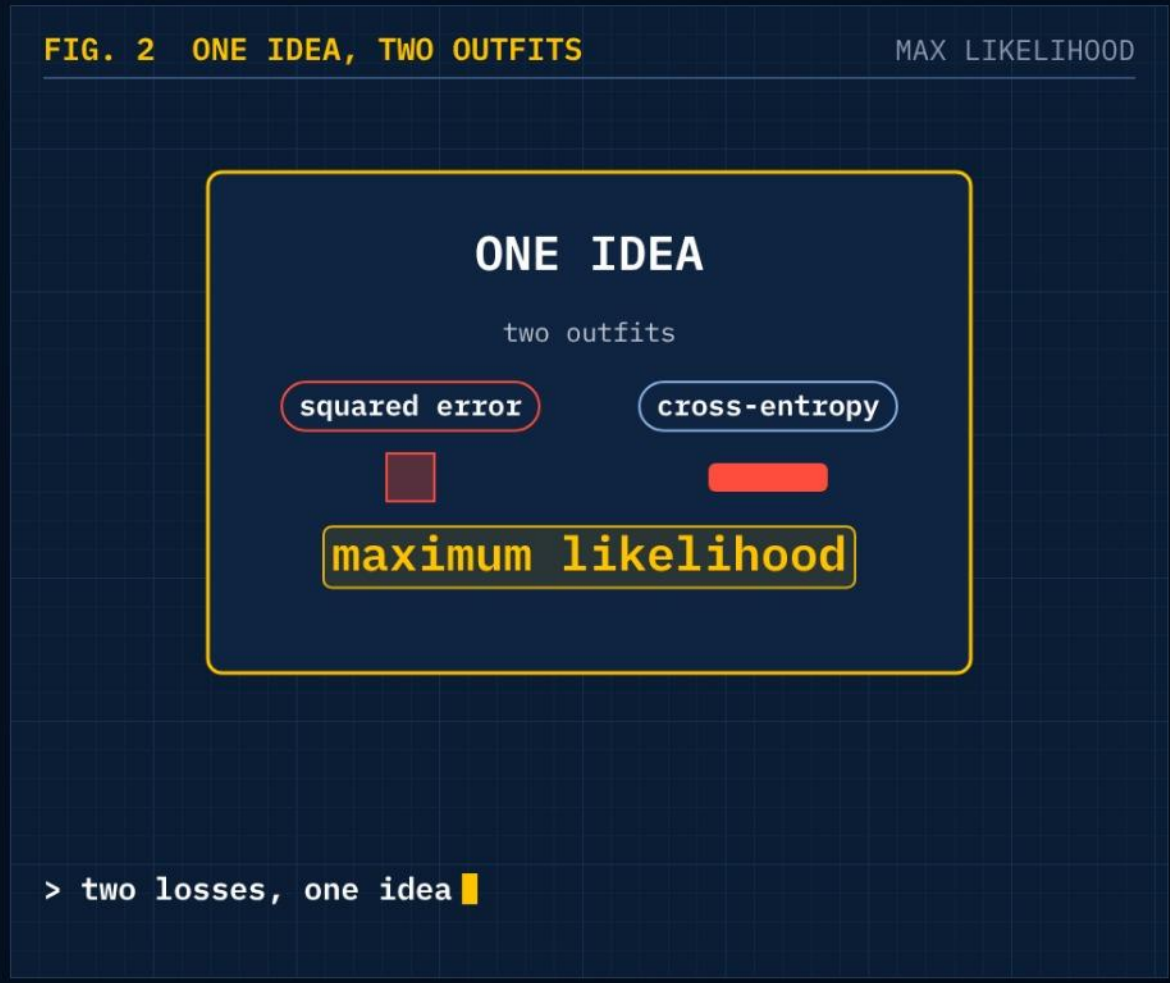
Two losses, one idea

Episode 7 met two losses. They looked like two separate choices.

01 Numbers: squared error. Why square?

02 Words: cross-entropy. Why a log?

03 One idea, two outfits: maximum likelihood



Episode 7 (the loss, by hand). Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 5 §5.5 (maximum likelihood estimation).

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EPISODE 8 · WHAT LEARNING MEANS, PART 2

Maximum likelihood

Three parts, every number worked by hand, and a notebook with all of them.

01 The likelihood: multiplying becomes averaging

02 The bell curve: where the square comes from

03 The fine print: assumptions, plus a myth



Companion notebook (Colab), linked in the description. Sources on every slide.

MARK

PART 1 · THE LIKELIHOOD

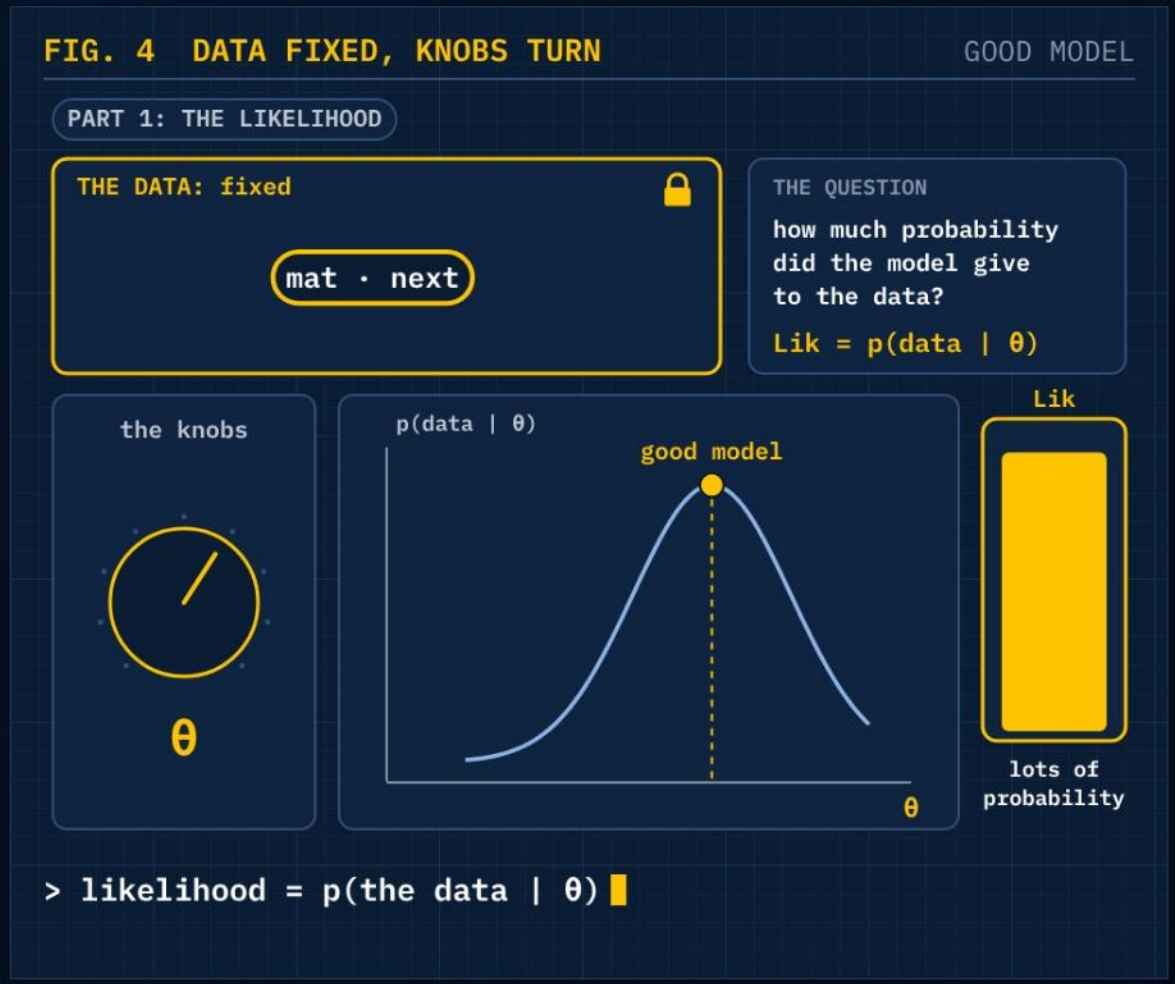
How probable was what happened?

The probability the model gives to the data that actually happened.

01 Of everything possible, the data happened

02 The data stay fixed; we turn θ

03 A good model gives the data a lot of probability



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 5 §5.5 (maximum likelihood estimation).

PART 1 · THE LIKELIHOOD

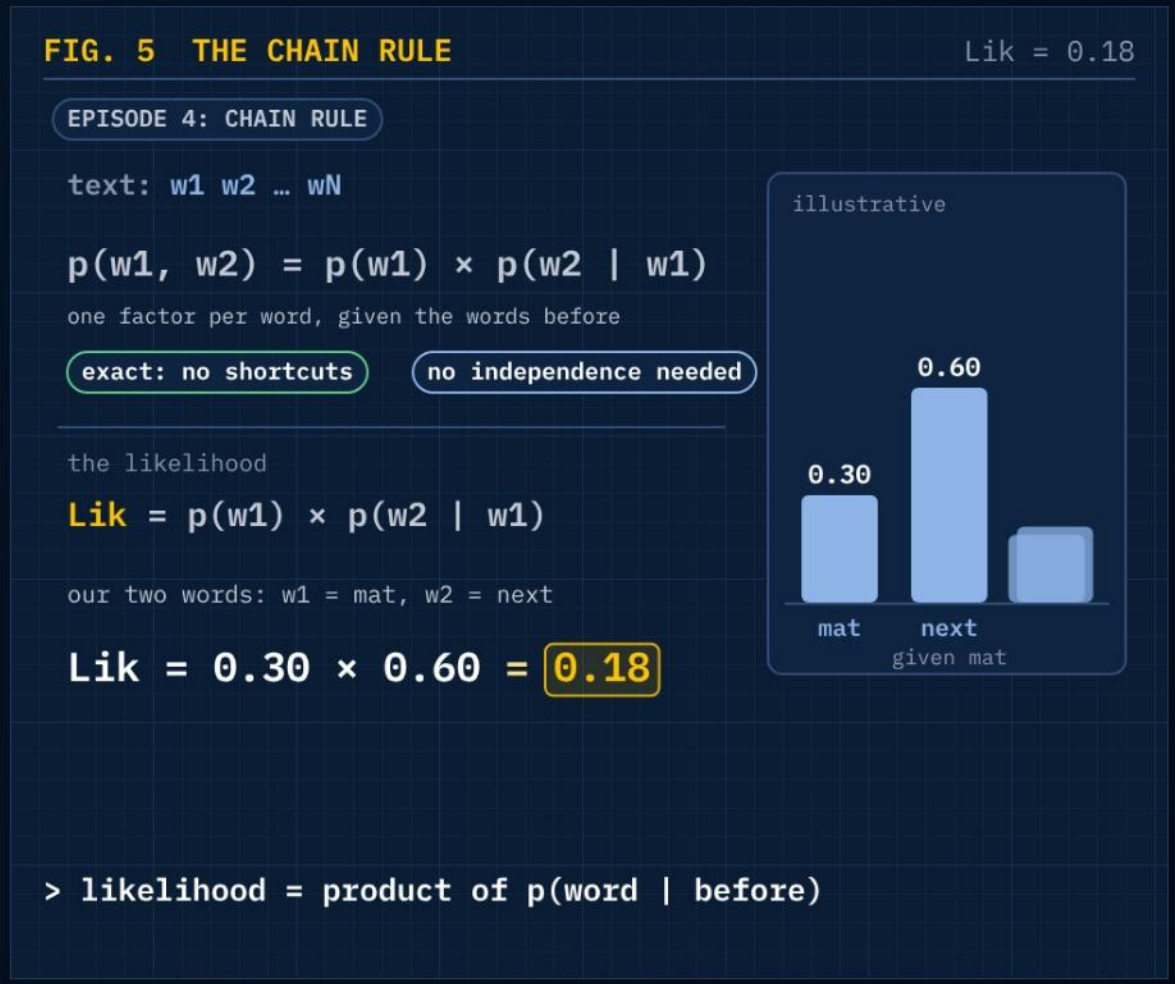
The chain rule does the work

$$p(w_1, w_2) = p(w_1) \times p(w_2 \mid w_1)$$

01 A sequence's probability: a product of next-word probabilities

02 Exact: no independence between words needed

03 Our two words: 0.30 × 0.60 = 0.18



Episode 4 (chain rule of probability). Jurafsky & Martin, *Speech and Language Processing*, 3rd ed. draft, ch. 3 (chain rule) and ch. 7. The word probabilities (0.30, 0.60; model 2: 0.40, 0.50) are illustrative, not measured.

PART 1 · TWO MODELS

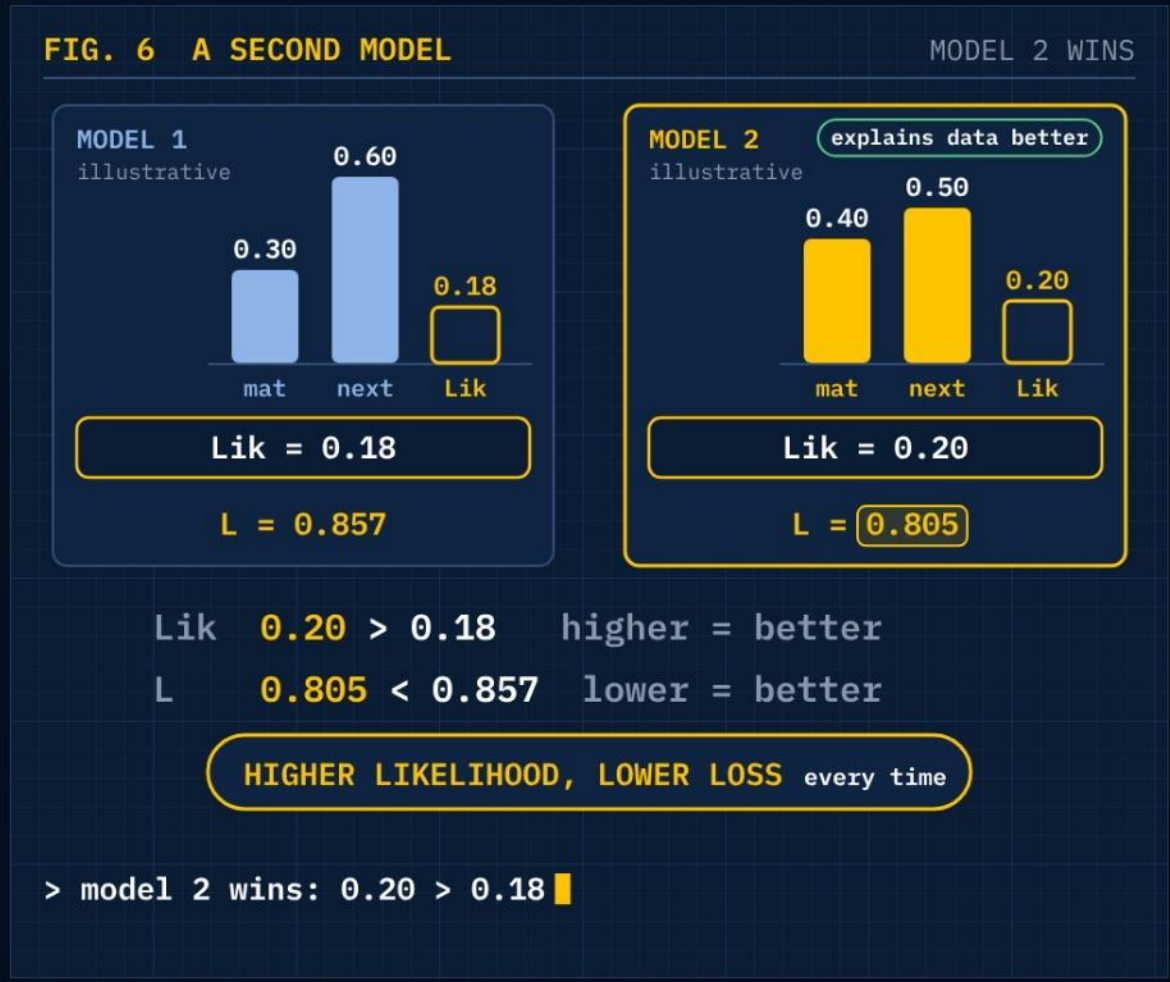
Higher likelihood, lower loss

Same two words, a second model.

01 Model 2: 0.40 × 0.50 = 0.20

02 0.20 > 0.18: model 2 explains the data better

03 Its loss: 0.805 < 0.857



Episode 7's check question. The word probabilities (0.30, 0.60; model 2: 0.40, 0.50) are illustrative, not measured.

PART 1 · THE LOG

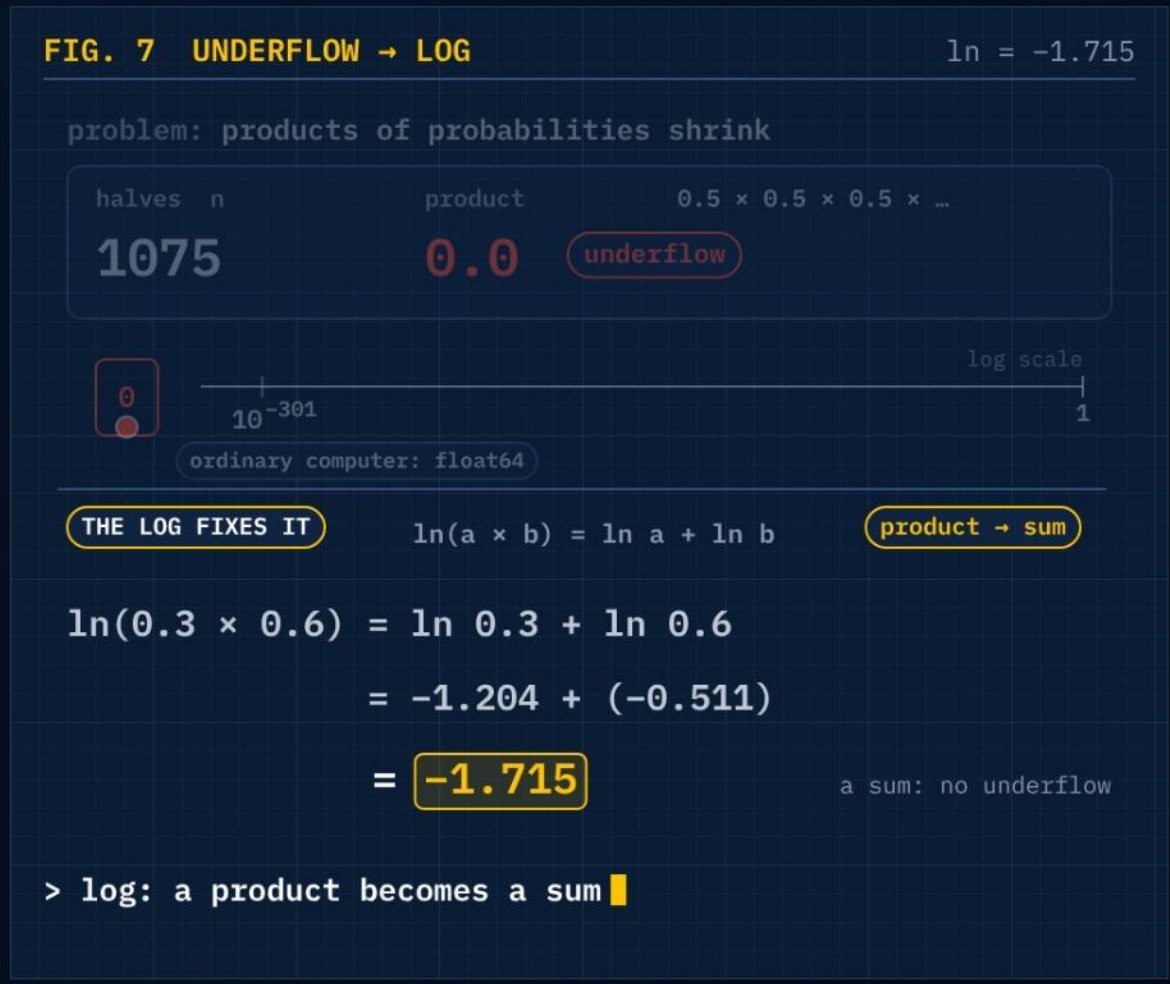
Products shrink. Logs add.

$$\ln(a \times b) = \ln a + \ln b$$

01 $0.5^{1000} \approx 10^{-301}$; 1,075 halves round to 0

02 The log turns the product into a sum

03 $\ln 0.3 + \ln 0.6 = -1.204 + (-0.511) = -1.715$



Underflow: IEEE 754 double precision (float64). Worked example; every number is checked in the companion notebook.

The log keeps the winner

The log only goes up: bigger in, bigger out.

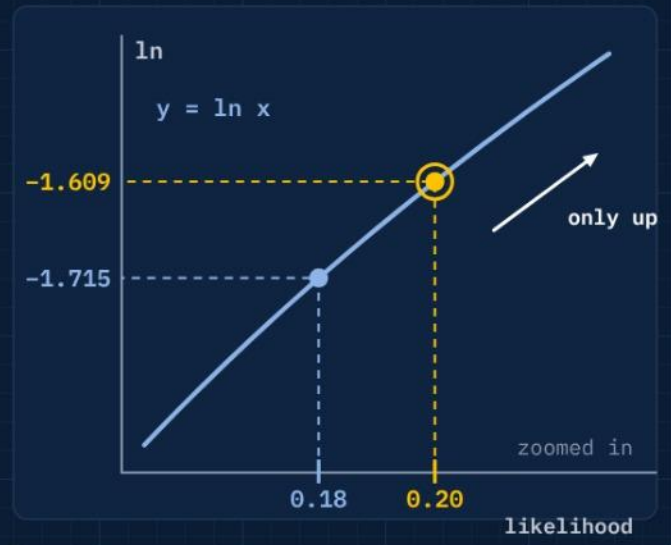
01 ln is monotonic: order is kept

02 ln 0.20 = −1.609 > ln 0.18 = −1.715

03 Taking the log “does not change its arg max”

FIG. 8 LOG KEEPS THE WINNER

WINNER: MODEL 2



does the log change who wins?

NO ✓

bigger in → bigger out

0.20 > 0.18

-1.609 > -1.715

max likelihood: model 2

max log-lik: model 2 ✓

GOODFELLOW, BENGIO & COURVILLE · DEEP LEARNING

“taking the logarithm of the likelihood does not change its arg max”

> log only goes up: same winner

Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 5 §5.5 (maximum likelihood estimation).

PART 1 · TWO MORE MOVES

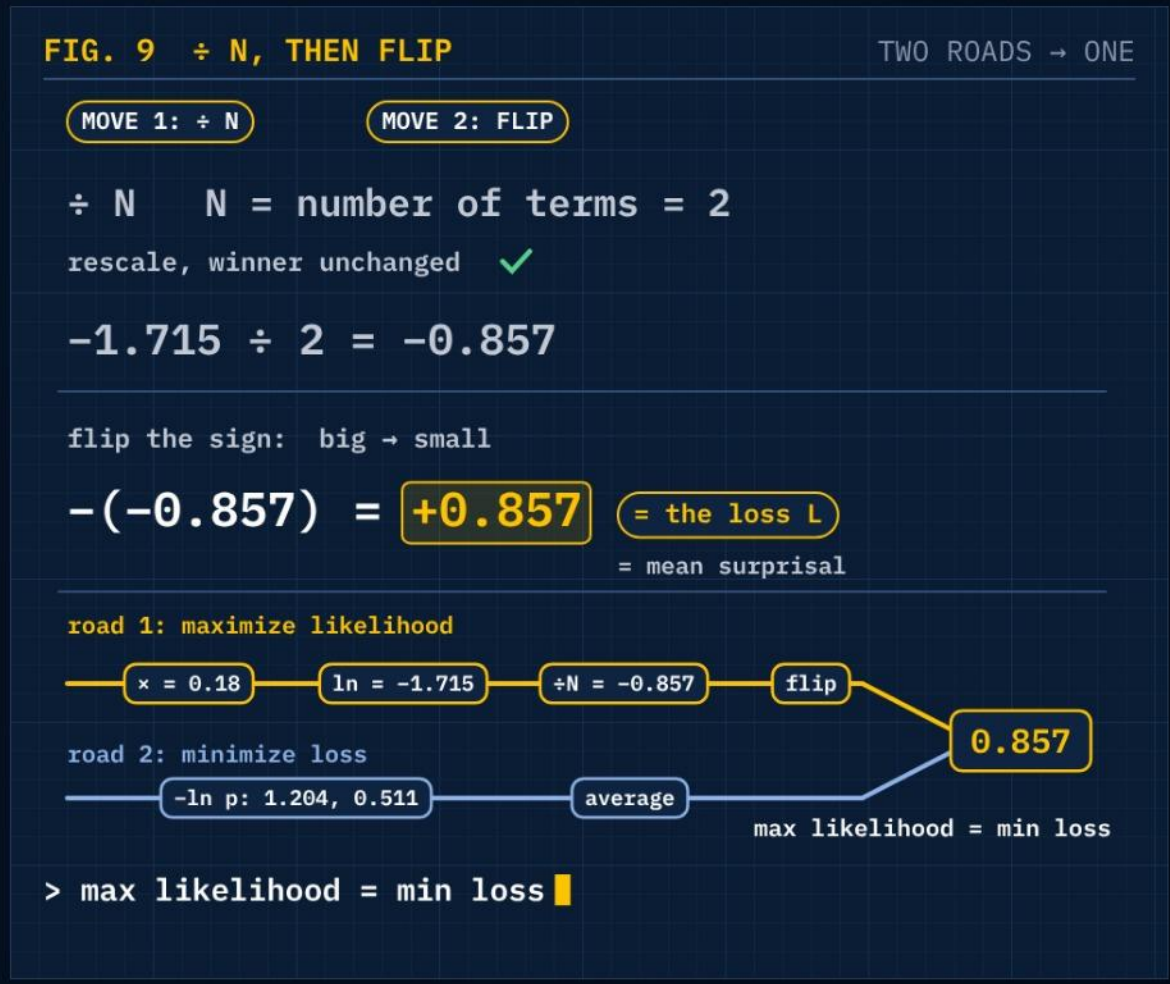
Divide by N, flip the sign

Rescaling doesn't change the winner.

01 −1.715 ÷ 2 = −0.857

02 Flip the sign: +0.857, the mean loss

03 Maximize likelihood = minimize the loss



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 5 §5.5 (maximum likelihood estimation). Worked example; every number is checked in the companion notebook.

PART 1 · THE PUNCHLINE

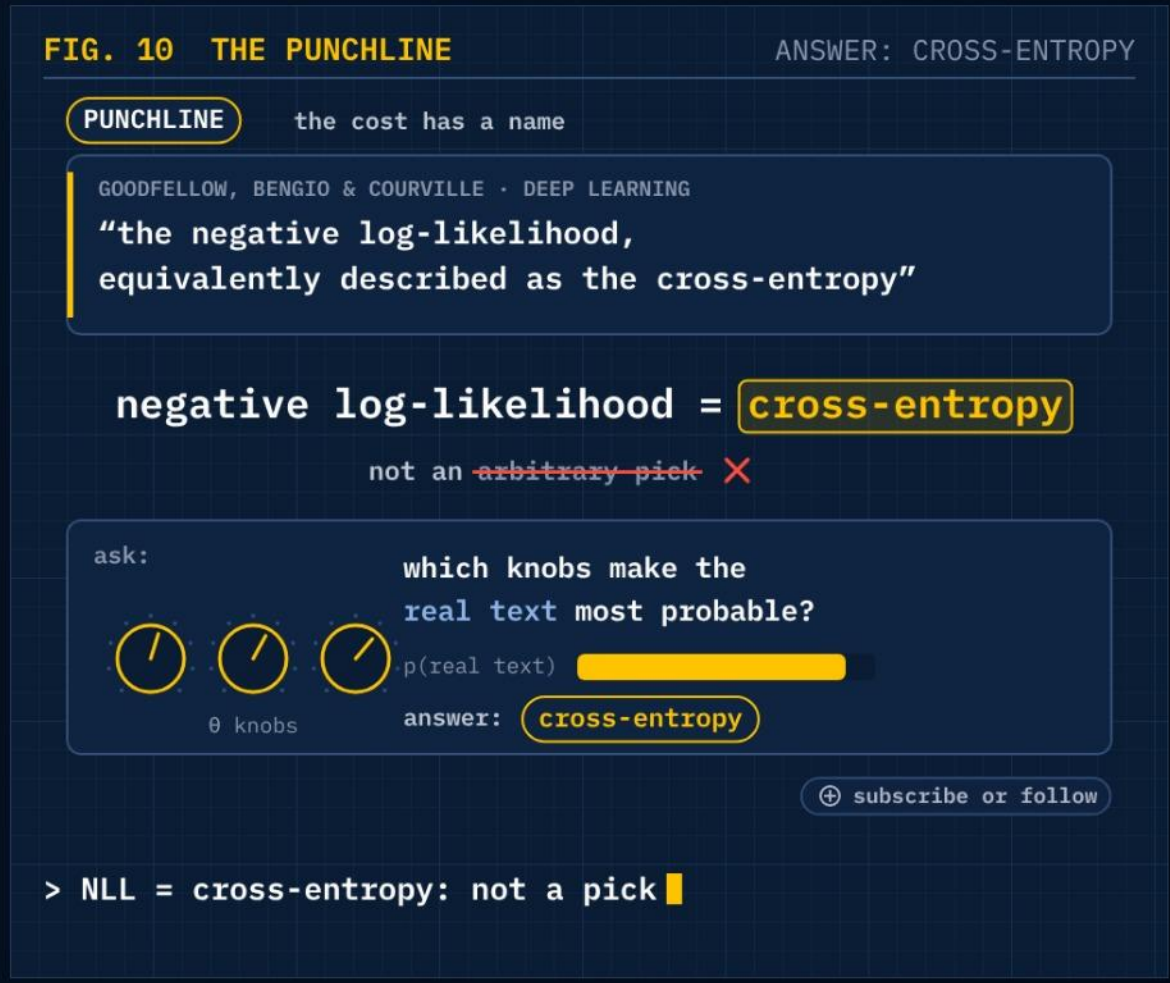
Negative log-likelihood is cross-entropy

"the negative log-likelihood, equivalently described as the cross-entropy"

01 Goodfellow, Bengio and Courville

02 Cross-entropy wasn't an arbitrary pick

03 Which knobs make the real text most probable?



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 6 §6.2.1.1 (learning conditional distributions with maximum likelihood).

PART 2 · NUMBERS

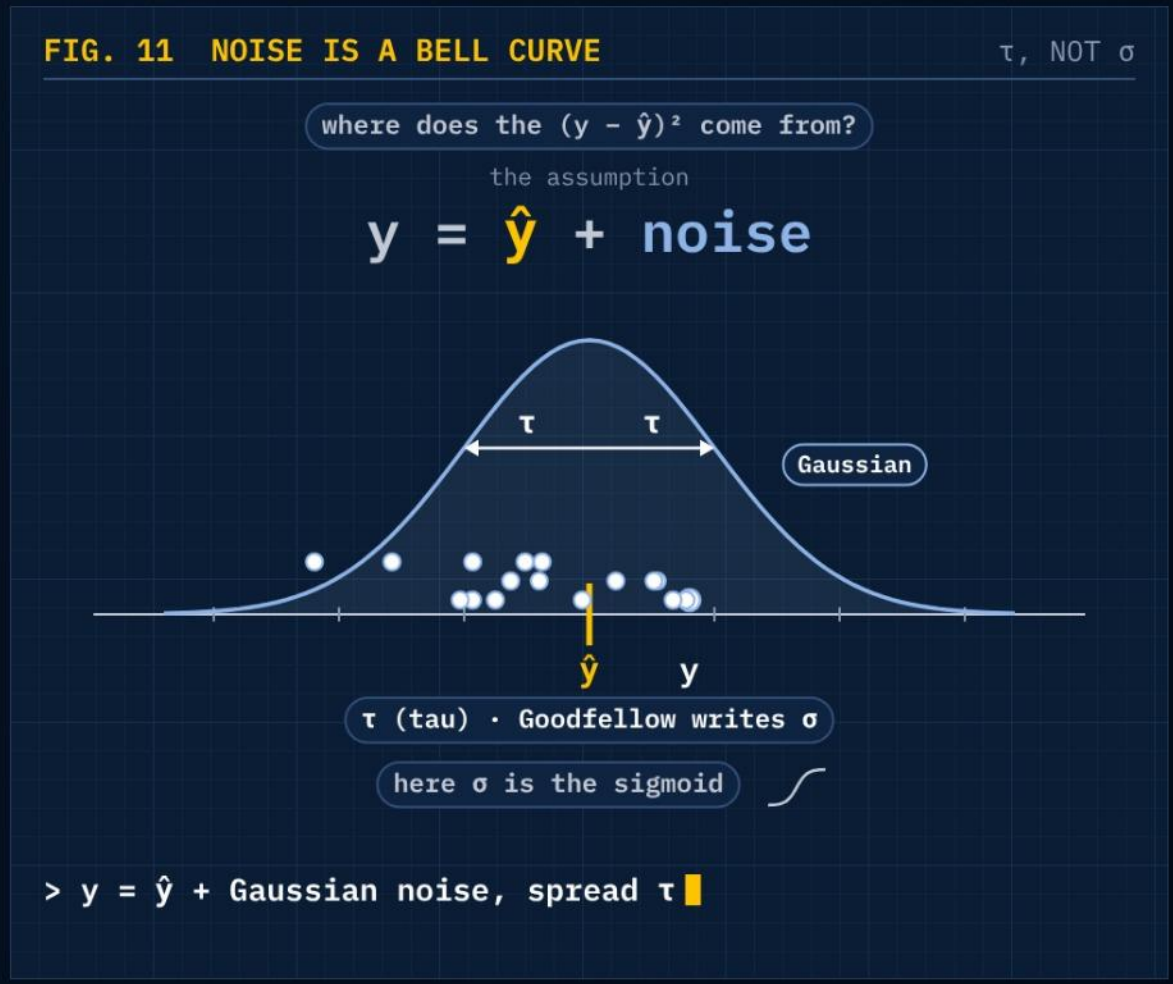
The assumption: Gaussian noise

y = ŷ + noise, and the noise follows a bell curve.

01 The label y = prediction ŷ plus noise

02 Noise: a Gaussian with a fixed spread τ

03 Goodfellow writes σ; here σ is the sigmoid



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 5 §5.5.1 (conditional log-likelihood and mean squared error, eq. 5.64–5.66).

PART 2 · THE BELL CURVE

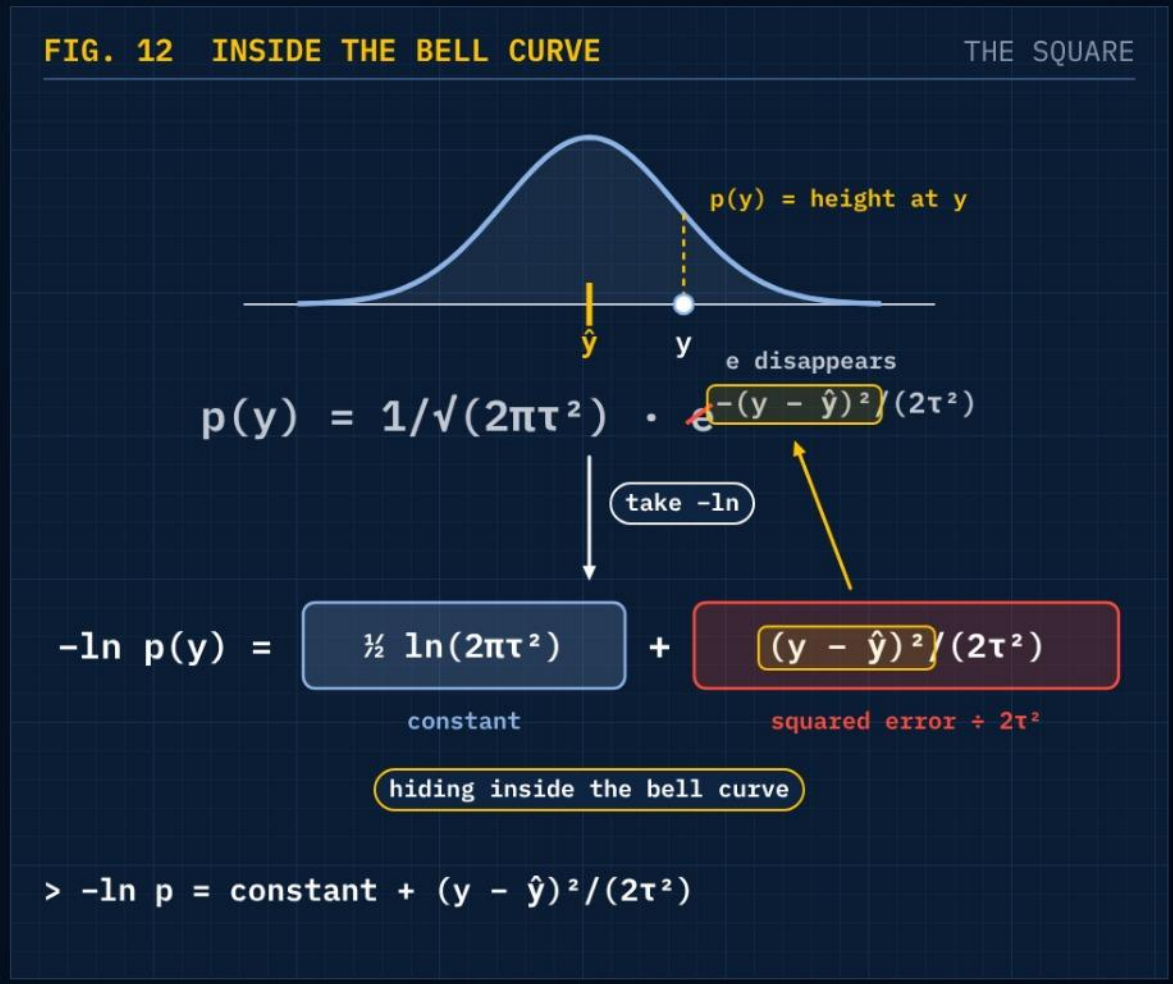
Where the square hides

The bell curve's height at y, then minus its log.

01 The bell curve's density at y

02 $-\ln p(y) = \tfrac{1}{2}\ln(2\pi\tau^2) + (y-\hat{y})^2/(2\tau^2)$

03 The square was inside the bell curve



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 5 §5.5.1 (conditional log-likelihood and mean squared error, eq. 5.64–5.66).

PART 2 · BY HAND

τ = 1, worked out

½ ln 2π: 2π ≈ 6.283, ln ≈ 1.838, half = 0.919

01 The constant: 0.919

02 Line A's miss of 1: 0.919 + 1²/2 = 1.419

03 A perfect hit pays just 0.919

FIG. 13 BY HAND, τ = 1 — HIT → 0.919

by hand, τ = 1

constant = ½ ln(2π)

2π ≈ 6.283

ln 6.283 ≈ 1.838

½ × 1.838 = 0.919

line A: miss of 1

0.919 + $1^2/2$

= 0.919 + 0.5

= **1.419**

perfect hit: 0.919 (just the constant)



> τ = 1: constant 0.919, miss 1 → 1.419

The three points are Episode 7's teaching example. Worked example; every number is checked in the companion notebook.

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PART 2 · ALL THREE POINTS

Same constant, different squares

Constant part: 3 × 0.919 = 2.757

01 Line A: 2.757 + ½ × 1 = 3.257

02 Line B: 2.757 + ½ × 1/6 = 2.840

03 Fixed τ: highest likelihood = least squares

FIG. 14 TOTALS FOR 3 POINTS

τ FIXED





A: 3 × 0.919 = 2.757
+ ½ × 1 = 3.257

B: same 2.757
+ ½ × 1/6 = 0.083
total = 2.840

fixed τ: highest likelihood = least-squares line

> max likelihood = least squares (τ fixed)

Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 5 §5.5.1 (conditional log-likelihood and mean squared error, eq. 5.64–5.66). The three points are Episode 7's teaching example. Worked example; every number is checked in the companion notebook.

MARK

PART 2 · SIDE BY SIDE

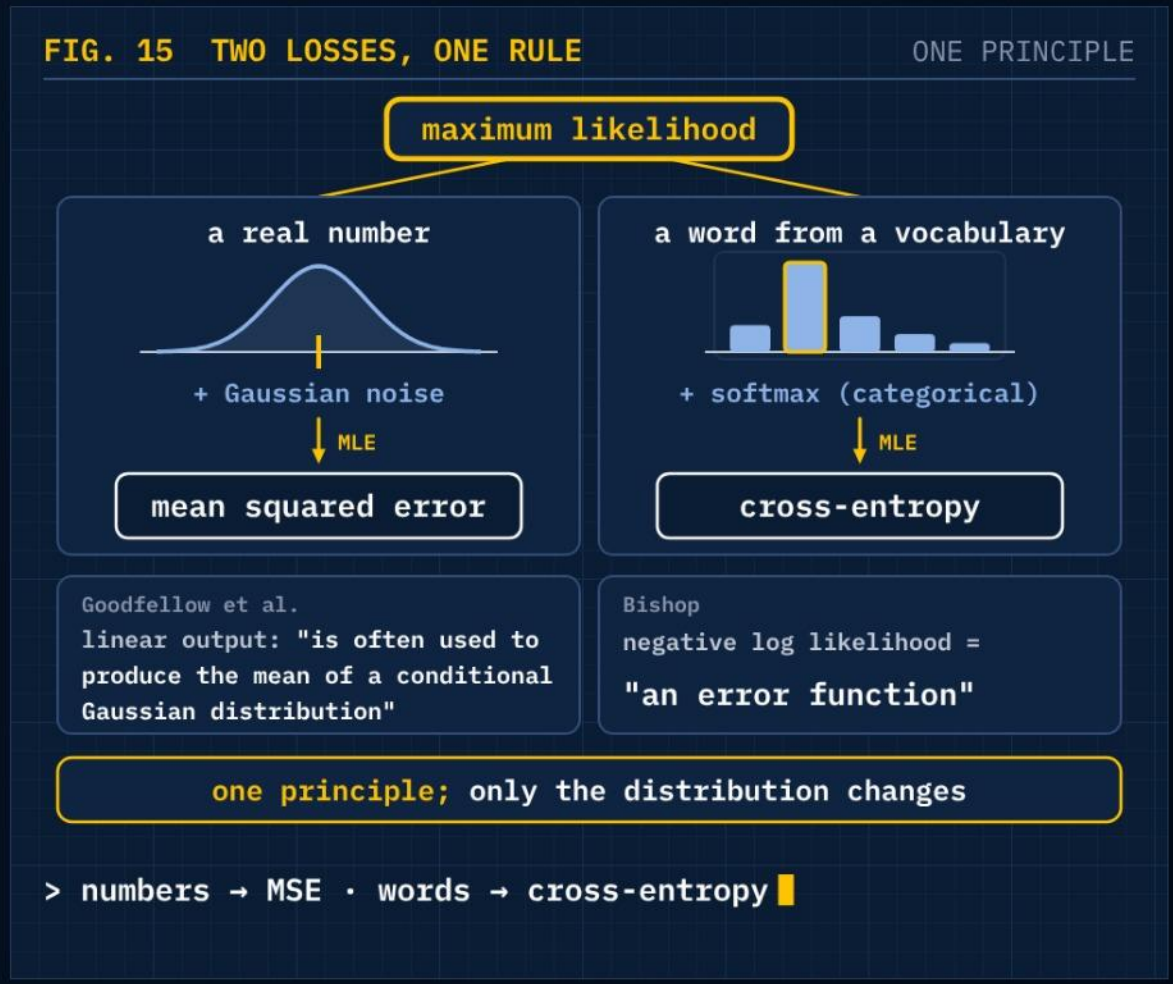
One principle, two losses

Only the assumed distribution changes.

01 A real number + Gaussian noise → mean squared error

02 A word + softmax (categorical) → cross-entropy

03 One principle: maximum likelihood



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 6 §6.2.2.1 (linear units for Gaussian output distributions). Bishop, *Pattern Recognition and Machine Learning* (Springer, 2006), §1.2.5 and §3.1.1.

PART 3 · THE FINE PRINT

Four assumptions

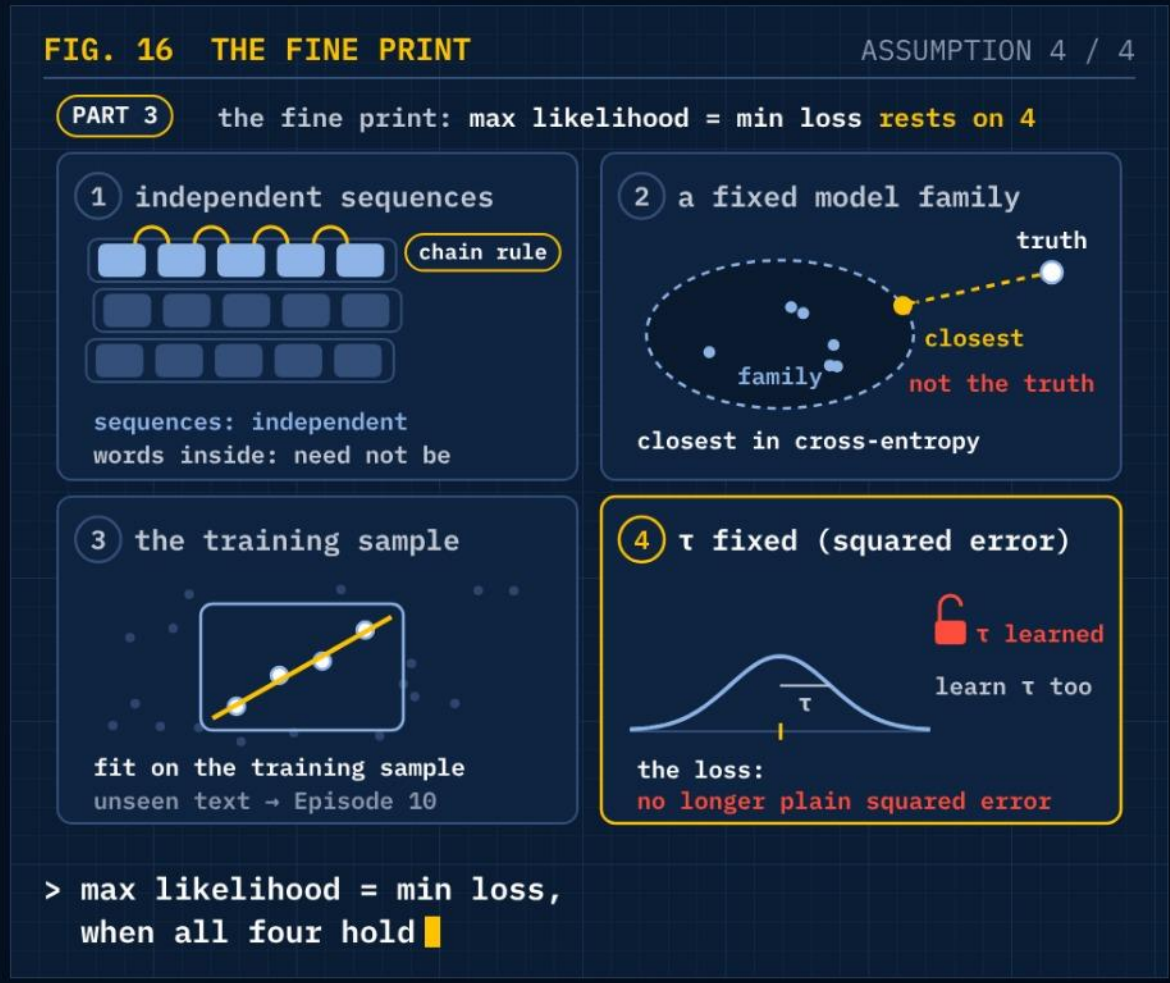
The equivalence rests on all four.

01 Training sequences drawn independently

02 A fixed model family

03 Fit on the training sample

04 For squared error: τ fixed



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 5 §5.5 and §5.5.1.

MYTH CHECK

“A low training loss means a good model”

Tempting: low loss means high likelihood.

01 The loss is averaged over the training data

02 A model can memorize and stumble on new text

03 Generalization: Episode 10



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 5 §5.2 (capacity, overfitting and underfitting).

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PART 3 · ONE MORE DETAIL

÷ N sets the gradient's size

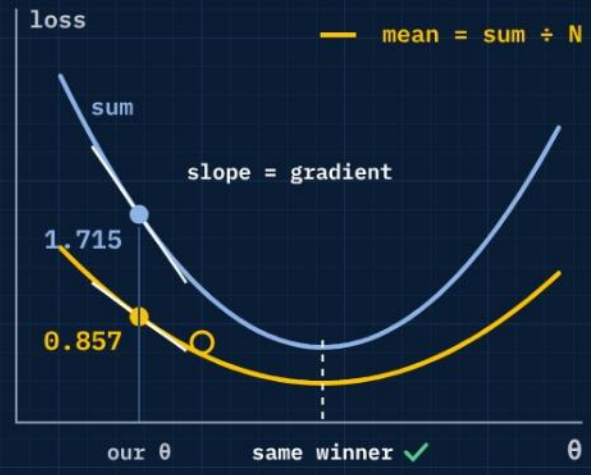
Same winner, different scale.

01 Sum 1.715 vs mean 0.857: gradient twice as big

02 Millions of words: millions of times bigger

03 So the step size η is set against the mean

FIG. 18 ÷ N AND THE GRADIENT → EPISODE 9



gradient size

sum 1.715

×2 (N = 2)

mean 0.857

N = millions

→ millions of times bigger

$$\theta \leftarrow \theta - \eta \cdot \text{gradient of the mean}$$

η = step size

→ Episode 9: the learning rate

> set η against the mean, not the sum

Episode 6 (the step) and Episode 9 (the learning rate). Worked example; every number is checked in the companion notebook.

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RECAP

Maximum likelihood, minimum loss

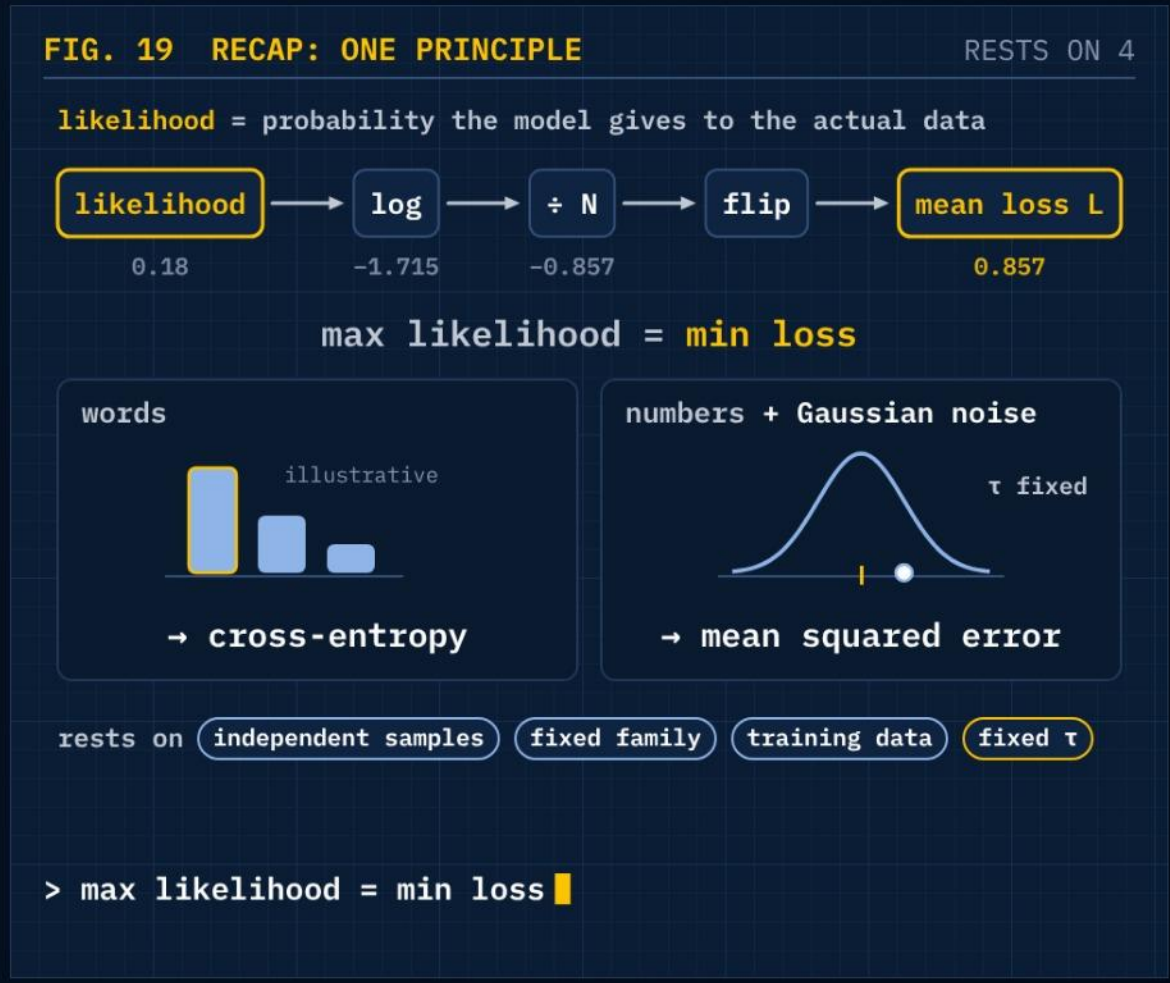
Likelihood: the probability the model gives to what actually happened.

01 log → ÷ N → flip = the mean loss

02 Maximize likelihood = minimize loss

03 Words: cross-entropy · numbers: MSE

04 Rests on four assumptions



All sources are listed in the description.

MARK

YOUR TURN

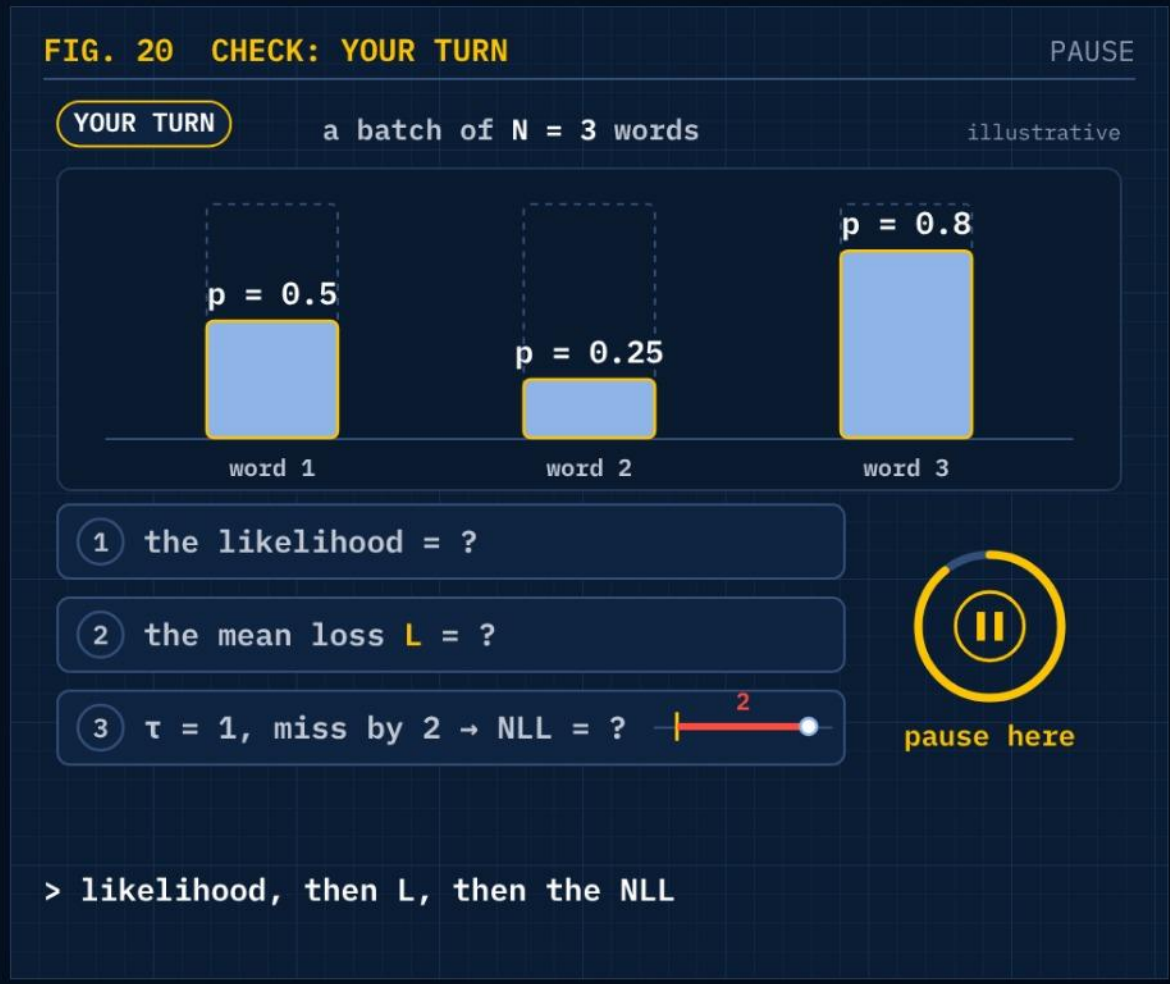
One question to take with you

Three words, observed at 0.5, 0.25 and 0.8.

01 The likelihood?

02 The mean loss L?

03 τ = 1, a miss by 2: its NLL?



Worked example; every number is checked in the companion notebook.

YOUR TURN

The answer

Probabilities → likelihood → loss.

01 0.5 × 0.25 × 0.8 = 0.1

02 −ln 0.1 = 2.303; ÷ 3 = 0.768 nats

03 0.919 + 2²/2 = 2.919

FIG. 21 ANSWER: BY HAND — ALL THREE

p = 0.5, 0.25, 0.8 N = 3

Q1 likelihood ✓

0.5 × 0.25 = 0.125

0.125 × 0.8 = **0.1**

Q2 mean loss L ✓

−ln 0.1 = 2.303

L = 2.303 ÷ 3 = **0.768** nats

Q3 τ = 1, miss by 2 ✓

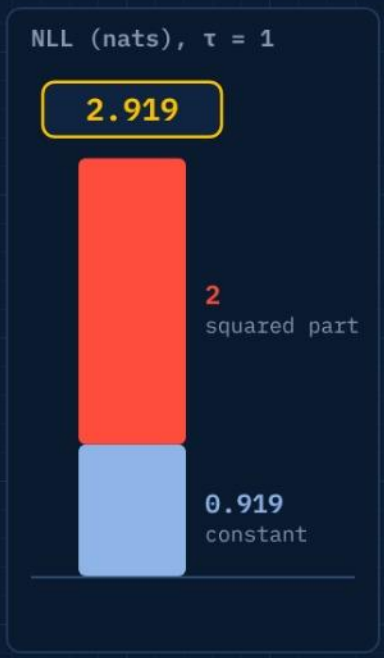
0.919 + 2²/2

= 0.919 + 2

= **2.919**

probabilities → likelihood

> 0.1 · 0.768 nats · 2.919



Worked example; every number is checked in the companion notebook.

TRY THIS WEEK

Three things to try.

01

Multiply three probabilities, then add their logs. Same answer both ways?

02

In the notebook, run the grid search: lowest loss, highest likelihood, same line

03

Work out the Gaussian NLL for line C (W 1.5, β 1/2)

THE MISTAKE TO AVOID

Treating low training loss as proof the model is good. It only says it fits what it has seen.

Companion notebook and every source in the description.

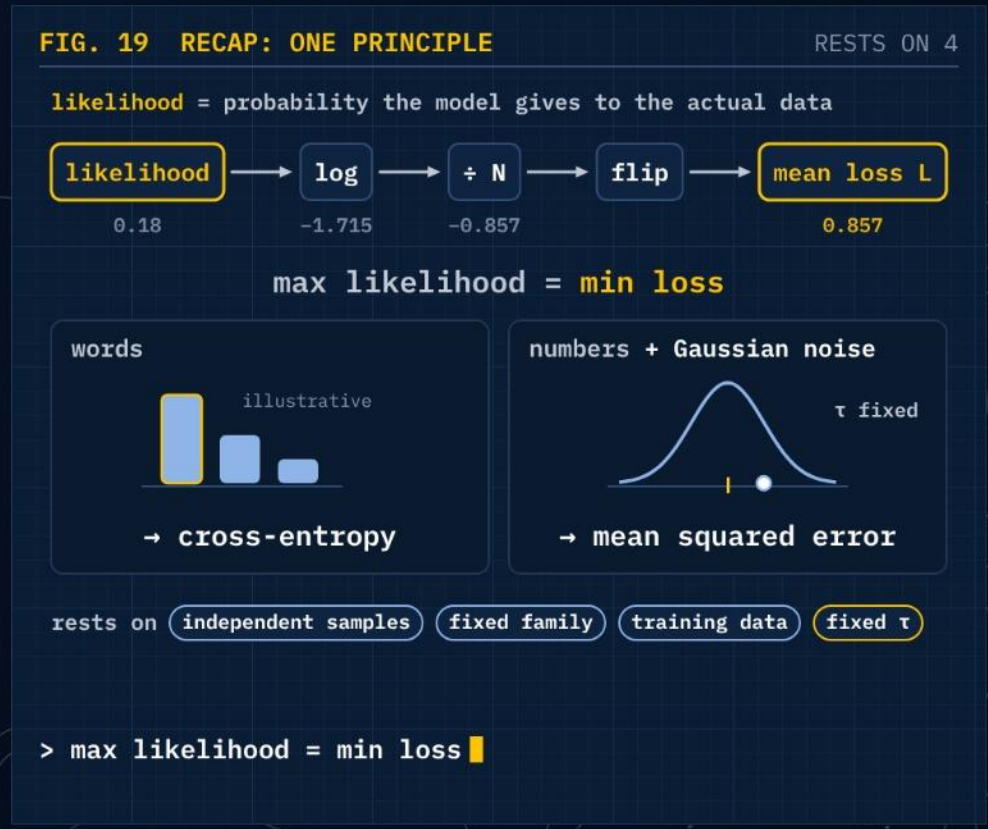
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NEXT TIME

Episode 9: Gradient descent, walking downhill for real, and the learning rate.

In the comments: did you see it coming that the two roads are one?

How Large Language Models Work · subscribe or follow for Episode 9



MARK